
Anonymous supplement: rational VP Heun certificate

Anonymous Author(s)

Affiliation

Address

email

1 This supplement records the integer comparison used in the workshop paper. It contains no author
2 names, emails, or public repository URLs.

3 The four-step VP Heun product is the following element $P(\pi, \sqrt{2})$ of $\mathbb{Q}[\pi, \sqrt{2}]$, obtained by ex-
4 panding the four Heun factors with $h = 1/4$ and reducing $\sqrt{2}^2 = 2$:

$$\begin{aligned} P(\pi, \sqrt{2}) = & 1 + \frac{3}{40}\pi + \frac{15}{164}\pi\sqrt{2} + \frac{78579}{10758400}\pi^2 + \frac{45}{5248}\pi^2\sqrt{2} \\ & + \frac{11421}{17213440}\pi^3 + \frac{4671}{17213440}\pi^3\sqrt{2} \\ & + \frac{5103}{275415040}\pi^4 + \frac{567}{55083008}\pi^4\sqrt{2} \\ & + \frac{243}{2203320320}\pi^5 + \frac{729}{2203320320}\pi^5\sqrt{2} + \frac{729}{176265625600}\pi^6. \end{aligned} \tag{1}$$

5 Each of the twelve coefficients is a ratio of nonnegative integers, hence nonnegative, and the constant
6 term is 1. Substituting $\pi < 355/113$ and $\sqrt{2} < 99/70$ is therefore valid and yields

$$P\left(\frac{355}{113}, \frac{99}{70}\right) = \frac{192113353671412470139303}{102753427879939708813312} < \frac{187}{100},$$

7 equivalently

$$19211335367141247013930300 < 19214891013548725548089344.$$

8 Hence $r_{\text{VP}} < 187/100$ and $W_2^{\text{VP}} > 13/100$. The bound $\sqrt{2} < 99/70$ is $99^2 - 2 \cdot 70^2 = 1$. The
9 bound $\pi < 355/113$ follows from Machin's formula with Leibniz remainders (seven Taylor terms
10 for $\arctan(1/5)$, four for $\arctan(1/239)$). An independent mpmath 1.3.0 interval enclosure is a
11 cross-check, not the proof.